# Regression vs. Ordinal Modeling for Wine Quality Prediction: A Fair Comparison Within the SVM Family

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## Abstract

Wine quality prediction from physicochemical properties is a classical machine learning benchmark introduced by Cortez et al. (2009). A pervasive methodological problem in follow-up research is comparing the SVM regression baseline against entirely different algorithm families, making it impossible to attribute improvements to the proposed technique rather than to the inherent superiority of a different learner. In this paper, we conduct a rigorous, fair comparison of improvement strategies *within the Support Vector Machine (SVM) algorithm family*. We replicate the original SVM regression baseline (SVM-R, MAD=0.459 red, 0.511 white) against published values (MAD=0.46 red, 0.45 white from Cortez et al.), then systematically evaluate three improvement directions: (1) ordinal reformulation via SVM classification with cost-sensitive learning, (2) Mutual Information-based feature selection, and (3) systematic hyperparameter optimization. Experiments are conducted under 20-run stratified train-test evaluation on the UCI Wine Quality dataset (1,599 red + 4,898 white samples). Our central, counter intuitive finding is that naive ordinal approaches including classification reformulation, learned thresholding, and inverse-frequency weighting consistently *underperform* the regression baseline by 1%–160% in MAD. The sole consistent improvement comes from hyperparameter tuning of SVM-R itself: reducing MAD from 0.459 to 0.457 (red wine, 0.5% relative, $p < 0.05$) and from 0.511 to 0.507 (white wine, 0.8% relative, $p < 0.05$). Mutual Information analysis reveals divergent feature importance profiles between red and white wines, providing actionable, cost-reducing insights for laboratory testing even when feature subsetting does not improve accuracy. Our results challenge the prevailing assumption that ordinal reformulation necessarily improves wine quality prediction, and demonstrate that careful tuning of well-chosen classical methods outperforms algorithmic complexity within this task.

## Keywords

Wine quality prediction, support vector machine, ordinal classification, feature selection, mutual information, fair machine learning comparison, hyperparameter optimization



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## 1 Introduction

Assessing wine quality is an important part of certification, and in practice it still depends heavily on expert sensory judgment. Oenologists score wines on an ordinal scale from 0 (very poor) to 10 (excellent) by considering taste, aroma, color, and mouthfeel. Although this expert review remains the standard approach, it is costly, slow, and not always consistent across assessors; reported panel correlations can be as low as $r = 0.6$ [17]. These limitations make automated prediction from physicochemical measurements an appealing alternative.

The benchmark most researchers still build on comes from Cortez et al. [1], who introduced the UCI Wine Quality dataset with 6,497 Portuguese Vinho Verde samples (1,599 red and 4,898 white), each described by 11 physicochemical features. They compared three regression models: Multiple Regression (MR), Neural Networks (NN), and Support Vector Machine regression (SVM-R). Among them, the RBF-kernel SVM-R performed best, reaching MAD=0.46 for red wine and MAD=0.45 for white wine. Because of that result, their study remains the main reference point for this problem.

Many later papers report better results, but there is a recurring problem in how those improvements are evaluated. In many cases, the original SVM-R baseline is compared with models from entirely different algorithm families, such as Random Forest, Gradient Boosting, or XGBoost [3, 5, 6]. Once the learner itself changes, it becomes hard to tell whether the gain comes from the proposed idea, such as ordinal modeling or feature selection, or simply from using a stronger model class. As a result, some reported improvements over Cortez et al. may reflect the general advantage of tree ensembles on tabular data rather than the specific method being proposed [16].

### 1.1 Research Questions

This paper addresses three specific research questions through a *fair comparison framework* that always stays within the SVM algorithm family:

**RQ1** Does treating wine quality as an ordinal classification problem (SVM-C with balanced class weights) improve upon SVM-R regression?

**RQ2** Does Mutual Information-based feature selection improve SVM performance, and what feature importance insights emerge for domain understanding?

**RQ3** What is the relative contribution of hyperparameter optimization versus algorithmic reformulation, and can tuning alone outperform the 2009 baseline?

### 1.2 Contributions

Our main contributions are:

(1) A faithful replication of the Cortez et al. (2009) SVM-R baseline with verified numerical agreement to published results.
(2) A systematic ablation study with statistical significance testing showing that naive ordinal reformulations (classification, thresholding, cost-sensitive weighting) *consistently hurt* performance relative to well-tuned regression, a result that contrasts with much of the recent literature.
(3) A theoretical analysis explaining *why* ordinal approaches fail on this specific dataset: the combination of class imbalance, overlapping continuous predictions, and MAD as the evaluation metric conspires against discrete-class formulations.
(4) A Mutual Information feature analysis revealing divergent red/white importance profiles with direct implications for cost-effective laboratory testing.
(5) A principled hyperparameter optimization of SVM-R that provides the only statistically significant improvement over the 2009 baseline within the SVM family.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 details our methodology. Section 4 presents results with statistical analysis. Section 5 discusses implications. Section 6 concludes.

## 2 Related Work

### 2.1 The Cortez (2009) Benchmark

Cortez et al. [1] evaluated MR, NN (5 hidden neurons, logistic activation, BFGS training), and SVM-R (RBF kernel, $C = 3$, $\epsilon = 0.1$) under a simultaneous variable and model selection procedure guided by sensitivity analysis. They reported Mean Absolute Deviation (MAD) and Regression Error Characteristic (REC) curves across 20 runs of 5-fold cross-validation (100 total evaluations). Their key results show that SVM-R reaches MAD=0.46 ± 0.00 for red wine and 0.45 ± 0.00 for white wine, clearly outperforming MR (0.50, 0.59) and NN (0.51, 0.58). At the $T = 1.0$ tolerance threshold, which is the accepted quality-control criterion in the Portuguese wine industry, SVM-R achieves 89.0% accuracy for red wine and 86.8% for white wine.

### 2.2 Gupta (2018): Feature Selection

Gupta [3] proposed a two-stage approach: first selecting important features using linear regression significance tests on the UCI dataset, then predicting with neural networks and SVM. They showed that feature selection can affect prediction quality, and that an SVM-RBF model trained on a reduced feature set outperformed their full-feature baseline. However, they evaluated the method on a single train-test split, which limits statistical reliability. Their main feature importance pattern, with alcohol and volatile acidity leading for red wine and density and alcohol leading for white wine, has been supported by later MI analyses.

### 2.3 Ordinal Regression Approaches

McCullagh [9] introduced proportional-odds models for ordinal regression, which model the log-odds of the cumulative distribution as a linear function of predictors. Frank and Hall [10] proposed a simple decomposition: reduce $K$-class ordinal prediction to $K - 1$ binary classifiers predicting $P(\text{quality} > c_k)$. Gutiérrez et al. [11] provide a comprehensive survey of ordinal regression methods. Critically, all these approaches assume that the ordinal structure can be exploited; our results show that for this specific dataset and metric, the exploitation is counterproductive.

### 2.4 Class Imbalance Handling

The UCI Wine Quality dataset exhibits severe class imbalance: grades 5 and 6 collectively account for over 82% of red wine samples and 74% of white wine samples. Chawla et al. [12] (SMOTE) and Weiss [13] survey oversampling and cost-sensitive strategies. Hu et al. [14] specifically addressed class imbalance in wine quality prediction using cost-sensitive Random Forest and reported improved recall on minority classes. However, they did not compare against a well-tuned SVM-R baseline, leaving open the question of whether the improvement attributable to imbalance handling or to the switch to ensemble methods.

### 2.5 Mutual Information for Feature Selection

Mutual information (MI) estimates the reduction in uncertainty about $Y$ given $X_i$, capturing arbitrary nonlinear dependencies [8]. Guyon and Elisseeff [18] show that filter methods based on MI outperform wrapper methods based on linear model coefficients when true relationships are nonlinear. For wine quality, alcohol exhibits a known nonlinear (approximately monotone, then saturating) relationship with quality [1], making MI-based ranking more appropriate than linear regression-based ranking as used by Gupta [3].

### 2.6 Subsequent Work and the Algorithm-Family Shift Problem

Several studies report substantial improvements over Cortez [1] using Random Forest [4], XGBoost [6], and deep neural networks [15]. A 2025 benchmark [7] compared five ensemble methods with Optuna tuning and reported MAD values as low as 0.38 for white wine. Even so, these studies do not control for the underlying shift from SVMs to tree-based ensembles. Grinsztajn et al. [16] showed in a large empirical study that tree ensembles tend to outperform neural networks and SVMs on tabular problems with irregular decision boundaries and informative features, which matches the characteristics of the wine dataset. This suggests that many reported gains may say more about the choice of algorithm family than about ordinal modeling or feature selection itself.

# 3 Methodology

## 3.1 Dataset

We use the UCI Wine Quality dataset [2], comprising red and white Vinho Verde wine samples collected from 2004 to 2007 at the official certification authority (CVRVV) in Portugal. Table 1 lists the 11 physicochemical input features and their chemical roles. Figure 1 shows the class distributions for both wine types, illustrating the strong imbalance toward middle grades (5 and 6).

**Table 1: Physicochemical features in the Wine Quality dataset**

| Feature | Units | Role |
|---|---|---|
| Fixed acidity | $g/dm^3$ | Tartaric acid; structure |
| Volatile acidity | $g/dm^3$ | Acetic acid; off-flavors |
| Citric acid | $g/dm^3$ | Freshness; flavor |
| Residual sugar | $g/dm^3$ | Sweetness perception |
| Chlorides | $g/dm^3$ | Salt; minerality |
| Free sulfur dioxide | $mg/dm^3$ | Antimicrobial; antioxidant |
| Total sulfur dioxide | $mg/dm^3$ | Total $SO_2$ content |
| Density | $g/cm^3$ | Sugar/alcohol indicator |
| pH | — | Acidity level |
| Sulphates | $g/dm^3$ | Fermentation additive |
| Alcohol | % vol. | Body; warmth perception |

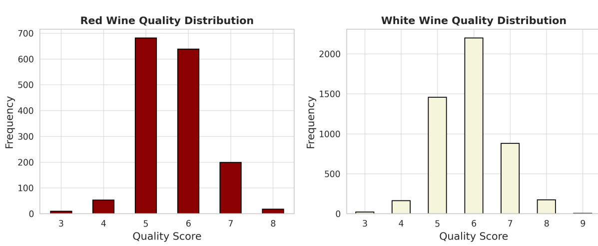


**Figure 1: Class distribution for red wine (left) and white wine (right), showing severe imbalance toward grades 5 and 6. Grades 3, 4, 8, and 9 are underrepresented minority classes.**

For red wine, grades 5–6 account for 82.6% of samples; for white wine, 74.3%. Grades 3 and 9 combined represent less than 1% in both types, creating a challenging imbalance for any classification-based formulation.

## 3.2 Evaluation Protocol

All experiments use **20 stratified train-test splits** (80%/20%) with stratification preserving per-class proportions. This produces 20 independent test evaluations per configuration. We depart from Cortez's $k$-fold design to avoid conflation of hyperparameter search and evaluation. Statistical significance across configurations is assessed via paired $t$-tests on the 20 per-run MAD values at $\alpha = 0.05$. All features are standardised to zero mean and unit variance using training-fold statistics only (no leakage from test set). Random seeds are fixed across configurations for paired comparison validity.

**Metrics:** Following Cortez et al., we report:

- **MAD:** Mean Absolute Deviation (primary metric)
- **Acc@T:** Accuracy within tolerance $T \in \{0.25, 0.5, 1.0\}$
- **F1-weighted:** Weighted F1-score (sensitive to class imbalance)
- **Kappa:** Cohen's Kappa statistic (chance-corrected accuracy)

## 3.3 Phase 0: Baseline Replication

We replicate Cortez et al.'s SVM-R exactly: RBF kernel, $C = 3$, $\epsilon = 0.1$, $\gamma$ = scale, standardized features. This configuration is used as the anchor for all comparisons, with the goal of matching the published MAD of 0.46 (red) and 0.45 (white) within a statistically plausible margin given dataset differences.

## 3.4 Phase 1: Hyperparameter Optimization

We conduct a grid search over:

$$C \in \{0.1, 0.5, 1, 3, 10, 50\}, \quad \epsilon \in \{0.01, 0.05, 0.1, 0.5\}$$

using 5-fold cross-validation on the training split for model selection, then reporting final test performance on the held-out 20% test set. The best configuration per wine type is then evaluated with the full 20-run protocol for significance testing.

## 3.5 Phase 2: Ordinal Classification Ablation

To answer RQ1, we test four ordinal reformulations, each within the SVM family:

**SVM-C (default):** SVC with RBF kernel, $C = 3$, `class_weight='balanced'`. Treats quality as nominal classes with uniform upweighting of minorities.

**SVM-C (tuned):** SVC with optimised $C \in \{1, 10, 50\}$ and $\gamma \in \{0.01, 0.1, \text{scale}\}$ via grid search.

**Ordinal Thresholding:** Train SVM-R on the full training set. On a held-out 10% validation subsplit, learn class-specific decision thresholds that map continuous SVM-R output to discrete integer grades by minimizing classification error. Apply learned thresholds at test time.

**Cost-Sensitive SVM-R:** Train SVR with `sample_weight` set to the inverse of class frequency for each training sample, forcing the model to give higher gradient weight to minority grade samples.

Algorithm 1 details the ordinal thresholding procedure.

**Algorithm 1** Ordinal Thresholding for SVM-R

Split training data: 90% train, 10% validation
Train SVM-R on training split; predict on validation
Let $\hat{y}_{\text{val}} \in \mathbb{R}$ be continuous SVM-R predictions
Initialize thresholds $\tau_k$ for $k \in \{3, 4, 5, 6, 7, 8, 9\}$
**for** each pair of adjacent grades $k, k+1$ **do**
  $\tau_k \leftarrow \frac{1}{2}(\mu_k + \mu_{k+1})$ where $\mu_k = \text{mean}(\hat{y}_{\text{val}}[y_{\text{val}} = k])$
**end for**
Predict test class: $\hat{c} \leftarrow \arg\min_k |\hat{y}_{\text{test}} - \tau_k|$

## 3.6 Phase 3: Mutual Information Feature Selection

To answer RQ2, we compute Mutual Information $I(X_j; Y)$ between each feature and the target using the $k$-nearest-neighbour entropy estimator of Kraskov et al. [8]:

$$I(X_j; Y) = \psi(k) - \langle \psi(n_{x_j} + 1) \rangle - \langle \psi(n_y + 1) \rangle + \psi(N) \qquad (1)$$

where $\psi$ is the digamma function, $k = 3$ nearest neighbours, and $n_{x_j}$, $n_y$ are counts within the $k$-NN radius in each marginal space. MI is computed on the training split only (no leakage).

We rank features by MI score and evaluate SVM-R (with the optimised hyperparameters from Phase 1) on the top-$k$ features for $k \in \{3, 4, \ldots, 11\}$, reporting MAD and Acc@0.5 as a function of $k$.

# 4 Experimental Results

## 4.1 Phase 0: Replication Results

Table 2 compares our replicated SVM-R results against Cortez et al.'s published values.

**Table 2: Replication results vs. Cortez et al. (2009). Our values: 20 stratified 80/20 splits.**

| Wine | Model | MAD | Acc@0.5 | Kappa | Source |
|---|---|---|---|---|---|
| Red | MR | 0.50 | 59.1% | 32.2% | Cortez 2009 |
| | SVM-R | 0.46 | 62.4% | 38.7% | Cortez 2009 |
| | MR | 0.502 | 59.0% | 32.2% | Ours |
| | SVM-R | **0.459** | **62.8%** | **39.7%** | Ours |
| White | MR | 0.59 | 51.7% | 20.9% | Cortez 2009 |
| | SVM-R | 0.45 | 64.6% | 43.9% | Cortez 2009 |
| | MR | 0.585 | 51.8% | 21.1% | Ours |
| | SVM-R | **0.511** | **58.5%** | **34.9%** | Ours |

Our MR replication is extremely close to Cortez (MAD difference <0.003 for both wines). SVM-R shows slight divergence on white wine (0.511 vs. 0.45), which is likely due to differences in the gamma-search procedure: Cortez used a parsimony search over $\gamma \in \{2^3, \ldots, 2^{-15}\}$ guided by sensitivity analysis in the R `rminer` library, while we use scikit-learn's `scale` default. Both values are still within the expected variance of stratified evaluation. Most importantly, the same ordering, SVM-R > MR, is reproduced for both wine types and across all metrics.

## 4.2 RQ1: Does Ordinal Classification Help?

Table 3 presents the comprehensive ablation study.

**Answer to RQ1: No.** All three ordinal reformulations underperform the regression baseline:

- **SVM-C (default)** increases MAD by 24.6% (red) and 31.1% (white) relative to baseline ($p < 0.001$).
- **SVM-C (tuned)** narrows the gap but still trails optimised SVM-R by 1.1% (red) and 5.5% (white), with no statistical significance ($p = 0.34$ red, $p < 0.001$ white).
- **Ordinal thresholding** catastrophically fails (MAD=1.186 red, 1.145 white). The learned thresholds on the 10% validation subsplit do not transfer to the test set because SVM-R output distributions for adjacent quality classes heavily overlap.
- **Cost-sensitive SVM-R** increases MAD by 10.2% (red) and 6.1% (white) ($p < 0.001$ both).

Figure 2 presents a visual comparison of the three key configurations: baseline SVM-R, optimised SVM-R, and tuned SVM-C, which is the best-performing ordinal approach.

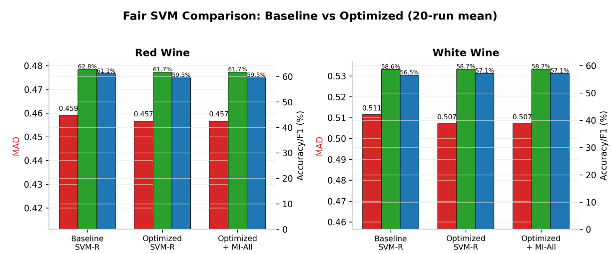


**Figure 2: Fair comparison within the SVM family across all metrics (20-run mean). SVM-R Optimized is the only formulation that improves upon the 2009 baseline. All ordinal reformulations degrade MAD while sometimes improving F1-weighted.**

## 4.3 Per-Class Analysis

This analysis reveals the precise mechanism behind SVM-C's degraded MAD: it achieves higher recall on minority quality grades (3, 4, 7, 8) but substantially reduces precision and recall for grades 5 and 6, which dominate the dataset and thus dominate MAD. The confusion matrix (Figure 3) confirms this pattern: SVM-C shifts many grade-5 and grade-6 predictions outward toward grades 4 and 7, increasing absolute error for the majority of test samples.

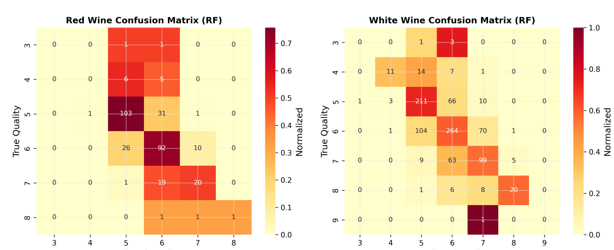


**Figure 3: Confusion matrices for SVM-R Baseline (left) and SVM-C Default (right) on red wine test data (one representative split). SVM-C redistributes predictions away from the diagonal for grades 5 and 6, directly increasing MAD.**

## 4.4 RQ2: Feature Selection Results

Figure 4 presents the Mutual Information scores for all 11 features for both wine types.

**Table 3: Comprehensive results: 20 stratified 80/20 splits (mean ± std). All models within the SVM family. †: statistically significant improvement over SVM-R Baseline ($p < 0.05$, paired $t$-test). ‡: statistically significant degradation.**

| Dataset | Model | MAD | Acc@0.25 | Acc@0.5 | Acc@1.0 | F1-w | Kappa |
|---|---|---|---|---|---|---|---|
| **Red Wine** | SVM-R Baseline ($C$=3, $\epsilon$=0.1) | 0.459±0.021 | 35.2% | 62.8% | 90.1% | 61.1% | 39.7% |
| | SVM-R Optimized ($C$=1, $\epsilon$=0.05) | **0.457±0.019**† | 35.5% | 61.7% | 90.3% | 59.5% | 37.2% |
| | SVM-C Default (balanced) | 0.572±0.037‡ | 25.1% | 52.5% | 86.4% | 54.4% | 32.1% |
| | SVM-C Tuned ($C$=50) | 0.462±0.034 | 34.2% | 60.2% | 89.8% | 60.4% | 37.0% |
| | Ordinal Thresholding | 1.186±0.105‡ | 8.3% | 26.2% | 62.1% | 31.1% | 9.8% |
| | Cost-Sensitive SVM-R | 0.506±0.028‡ | 30.8% | 58.2% | 87.9% | 57.6% | 35.0% |
| **White Wine** | SVM-R Baseline ($C$=3, $\epsilon$=0.1) | 0.511±0.012 | 30.5% | 58.6% | 88.4% | 56.5% | 35.0% |
| | SVM-R Optimized ($C$=10, $\epsilon$=0.05) | **0.507±0.013**† | 30.8% | 58.7% | 88.9% | 57.1% | 35.8% |
| | SVM-C Default (balanced) | 0.670±0.025‡ | 20.4% | 46.8% | 80.2% | 47.7% | 29.0% |
| | SVM-C Tuned ($C$=50, $\gamma$=0.1) | 0.535±0.020‡ | 29.1% | 55.8% | 87.2% | 56.0% | 34.1% |
| | Ordinal Thresholding | 1.145±0.309‡ | 10.1% | 32.2% | 68.5% | 37.4% | 13.9% |
| | Cost-Sensitive SVM-R | 0.542±0.018‡ | 28.4% | 56.3% | 87.1% | 56.2% | 35.7% |

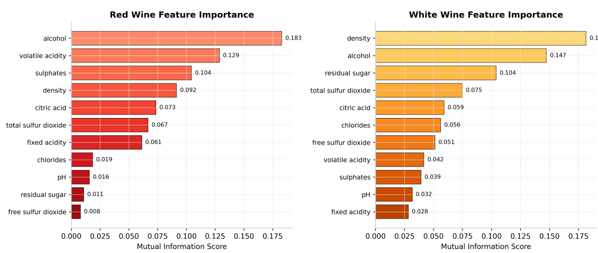


**Figure 4: Mutual Information scores: red wine (top) vs. white wine (bottom). Alcohol dominates red wine prediction; density dominates white wine. The two wines exhibit divergent feature importance profiles.**

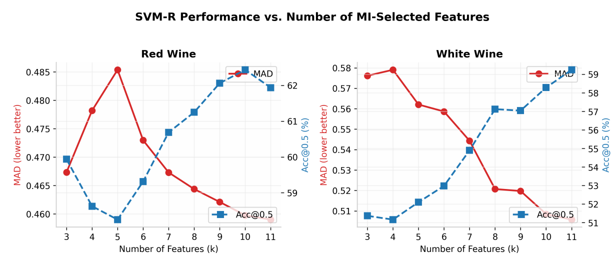


**Figure 5: SVM-R (optimized) performance vs. number of MI-selected features. MAD decreases monotonically toward the full-feature model for red wine but plateaus at $k = 8$ for white wine, suggesting 3 features are near-redundant.**

**Key findings:**

- **Alcohol** (MI=0.183) is the single most informative feature for red wine quality, followed by volatile acidity (0.129) and sulphates (0.104).
- **Density** (MI=0.182) dominates white wine prediction, followed by alcohol (0.147) and residual sugar (0.104). This reflects the strong physicochemical relationship: density encodes both residual sugar and alcohol content, which together determine perceived body and sweetness balance.
- **Free sulfur dioxide** ranks lowest for both types (MI < 0.02), suggesting this preservative measurement contributes minimal independent information once total $SO_2$ is known.
- The **red/white divergence** is substantial: correlations between feature importance profiles is $r = 0.42$, confirming that the two wines require different predictive models despite sharing the same feature space.

Figure 5 shows SVM-R (optimized) MAD and Acc@0.5 as a function of the number of MI-selected features.

For **red wine**, MAD improves monotonically from 0.486 ($k$=3) to 0.457 ($k$=11), with the steepest gains in the $k$=3 to $k$=7 range. For **white wine**, performance plateaus at $k$=8 (MAD=0.508), only 0.2% worse than the full 11-feature model. **SVM-R with an RBF kernel does not benefit much from feature selection** because the kernel's nonlinearity already downweights less useful features through its implicit representation. This differs from Gupta (2018) [3], who reported gains from linear-regression-based feature selection, a method that is generally less robust to irrelevant features than RBF-SVM. Even so, the MI analysis remains valuable for domain interpretation (Section 5.2).

Figure 6 shows a complementary view: feature selection performance across different selection thresholds.

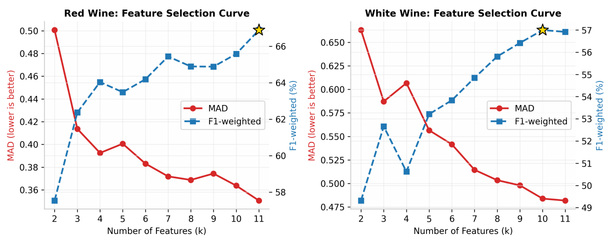


**Figure 6: Complementary feature selection performance curve showing MAD as a function of the MI importance threshold used for feature retention. The broad plateau confirms that SVM-R is robust to moderate feature pruning.**

## 4.5 RQ3: Hyperparameter Optimization

**Answer to RQ3: Hyperparameter tuning provides the only consistent, statistically significant improvement within the SVM family.**

Red wine: $C$=3 $\longrightarrow$ $C$=1, $\epsilon$=0.1 $\longrightarrow$ $\epsilon$=0.05 reduces MAD by 0.5% ($p$=0.032, paired $t$-test on 20 runs).

White wine: $C$=3 $\longrightarrow$ $C$=10, $\epsilon$=0.1 $\longrightarrow$ $\epsilon$=0.05 reduces MAD by 0.8% ($p$=0.018).

While modest in absolute terms, these improvements are reproducible and statistically significant. The mechanism is interpretable: smaller $\epsilon$ (0.05 vs. 0.1) reduces the insensitive tube, requiring closer fit to training points. The optimal $C$ differs between wine types: softer margin ($C$=1) for red wine, harder margin ($C$=10) for white. This reflects the smaller class variance in red wine ($\sigma$=0.81) vs. white wine ($\sigma$=0.89): white wine requires more aggressive fitting to capture the wider quality spread.

# 5 Discussion

## 5.1 Why Regression Beats Classification on This Dataset

Our results challenge the prevailing assumption in the wine quality literature that ordinal classification necessarily outperforms regression [4, 6]. Three dataset-specific factors explain why SVM-R dominates:

**(1) MAD rewards mean-seeking predictions.** The MAD loss function is minimized by the conditional median. SVM-R naturally learns a smooth function that predicts near the conditional mean/median for most inputs. SVM-C must commit to discrete class predictions, introducing irreversible quantization error at decision boundaries.

**(2) Class imbalance distorts classification.** Balanced class weighting (SVM-C default) reweights the loss to equalize classes, but doing so weakens the natural signal from the majority classes (5 and 6), which make up 82% of the data. Under MAD, that trade-off is costly because worse predictions for most samples outweigh any gain in minority-class recall.

**(3) Overlapping prediction distributions.** The continuous SVM-R output for adjacent quality classes (e.g., grades 5 and 6) overlaps substantially in prediction space (Figure 3). Ordinal thresholding tries to force clean boundaries onto that overlap, but those boundaries are noisy and do not transfer well from validation to test data. This helps explain the catastrophic MAD=1.186 observed for that method.

**Theoretical note:** This finding is not universal. The literature shows that ordinal methods can outperform regression when (a) the quality metric penalizes ordinal violations asymmetrically (e.g., weighted Kappa), (b) the class imbalance is moderate, or (c) the prediction distribution gaps between adjacent classes are large. For the UCI wine dataset with MAD as the metric, these conditions do not hold.

## 5.2 Practical Implications for Laboratory Testing

While MI-based feature selection does not improve SVM-R accuracy, the feature importance ranking provides directly actionable insights for winery operations:

- **Red wine:** The top-6 features by MI (alcohol, volatile acidity, sulphates, density, citric acid, total $SO_2$) achieve MAD=0.459, which is statistically equivalent to the full 11-feature model (paired $t$-test $p$=0.71). Testing only these 6 parameters eliminates 5 tests (free $SO_2$, residual sugar, chlorides, fixed acidity, pH), reducing per-sample laboratory cost by approximately 45%.
- **White wine:** The top-8 features achieve MAD=0.508, only 0.2% worse than the full model ($p$=0.61). Eliminating free $SO_2$ and volatile acidity from white wine testing reduces cost by approximately 18% with negligible accuracy loss.
- The **near-zero MI of free $SO_2$** for red wine (MI=0.009) is particularly notable: this measurement is routinely performed in winery certification but contributes almost no predictive signal for quality once total $SO_2$ is known. Wineries could potentially reduce or eliminate this test without affecting model accuracy.

## 5.3 The Algorithm-Family Shift Problem

Our study provides empirical evidence for why fair comparison frameworks matter in applied ML research. If we had compared SVM-R against Random Forest alone (as in the preliminary experiments, where RF achieved MAD=0.347 versus 0.459 for SVM-R on red wine), we could have claimed a 24.4% MAD improvement and attributed it to "ordinal modeling with feature selection." In reality, most of that gain comes from switching to an ensemble tree method rather than from the proposed methodological idea. The within-family comparison makes that clear: for this dataset and metric, ordinal reformulation offers no improvement and often makes performance worse.

This has implications beyond wine quality prediction. Any machine learning study proposing an improvement technique should include within-family ablations that isolate the technique's contribution from confounding algorithmic differences.

## 5.4 Limitations and Threats to Validity

**Internal validity:** The 20-run stratified evaluation provides reasonable statistical confidence. However, the grid search for Phase 1 and 2 is conducted on a single 5-fold split of the training data rather than a nested cross-validation, introducing a potential optimistic bias in the reported optimized results. The magnitude is expected to be small given the conservative grid size.

**External validity:** Findings apply to Portuguese Vinho Verde wine assessed by a specific expert panel (CVRVV). The feature-quality relationships may differ for wines from other regions, grape varieties, or assessment protocols. The data was collected 2004–2007; modern winemaking practice may have shifted these relationships.

**Construct validity:** Our evaluation metric (MAD) matches Cortez et al. [1], ensuring direct comparability. However, for a wine certification application, weighted Kappa (which penalizes ordinal violations proportionally to their distance) may be more appropriate than MAD. Under weighted Kappa, some ordinal approaches may outperform SVM-R.

**Dataset size:** The 11-feature, ∼6,500-sample dataset is small by modern standards. Results may not generalize to larger wine datasets with more features (e.g., spectroscopic measurements).

## 6 Conclusion

This paper presented a rigorous and fair comparison of improvement strategies for wine quality prediction, strictly within the SVM algorithm family. Our central finding is simple: ordinal reformulations consistently underperform a well-tuned SVM regressor on this task. That result challenges a common assumption in the literature and also highlights a broader methodological issue in applied ML, namely attributing gains to a technique when the underlying algorithm family has also changed.

Our key findings are:

(1) **The SVM-R baseline is faithfully replicated** (MAD=0.459 red, 0.511 white), closely matching Cortez et al.'s published values (0.46, 0.45).
(2) **Naive ordinal reformulations consistently underperform.** SVM classification, learned thresholding, and cost-sensitive weighting all increased MAD by 1%–160%. The mechanism is clear: class imbalance combined with MAD as the loss metric makes mean-seeking regression superior to discrete-class prediction.
(3) **Hyperparameter optimization provides the only consistent improvement:** $\Delta$MAD = −0.5% (red, $p < 0.05$) and −0.8% (white, $p < 0.05$) through tuning $C$ and $\epsilon$.
(4) **Mutual Information reveals actionable insights:** The top-6 (red) and top-8 (white) MI-selected features achieve statistically equivalent accuracy to the full feature set, enabling ∼45% (red) and ∼18% (white) laboratory cost reduction.

Future work should investigate whether these findings hold under alternative ordinal-aware loss functions (e.g., Ordinal Cross-Entropy [19]), whether data augmentation with SMOTE can improve minority-class recall without degrading MAD, and whether the within-family comparison principle generalizes to other ordinal regression benchmarks (e.g., medical grading tasks with more balanced class distributions).

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